

Nate Cirino

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Graduating May 2027 | Available for a 2027 start

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Algorithms & Data Structures, Computer Systems, Digital Design & Computer Architecture, Computer Networks, Foundations of Data Science, Object-Oriented Design, Foundations of Software Engineering, Systems Security

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair retail brands: e-labels, mobile, and customer-facing POS
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Integrated Expert AI (LLM-backed clinical decision support) into CME microservices, serving 250K+ clinicians
- Analyzed serialization bottleneck in Relay JMS broker and overhauled it with json-io across 20+ microservices, eliminating request timeouts under production load
- Optimized Expert AI retrieval, search, and filtering across 20+ microservices, improving query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing inference workflow
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Maintained 99.99% uptime across 1–2M monthly model inference calls for 2K paying users via AWS WAF, Datadog, and K8s autoscaling
- Architected Flask serving backend with RAG pipeline delivering personalized workout recommendations via Bedrock KB
- Evolved retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, improving recall and cutting serving overhead
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models in PyTorch on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined model architecture and loss formulation to address class imbalance and sensor noise, improving detection
- Built AWS Neuron SDK deployment path compiling models for onboard custom-silicon accelerator inference under hard real-time latency budgets

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes across a multi-node cluster; storage path: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Analyzed write-path bottlenecks and built single-tier compaction with full SSTable rewrite every 4 flushes

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via LLM tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems, using prompt caching on frozen schema prefixes to cut repeated prefill work; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

SKILLS & HONORS

Languages/Tools: Python, C++, C, Rust, Java, Bash, TypeScript, JavaScript, Linux, AWS, Azure, Kubernetes, Docker, Terraform, PostgreSQL, Jenkins, Buildkite, Groovy, GraphQL

Frameworks & Libraries: PyTorch, TensorFlow, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: json-io OSS Contributor, Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks